7.4.9

$$A = \begin{bmatrix} 3 & -3 \\ 0 & 0 \\ 1 & 1 \end{bmatrix}$$

Find a SVD of this.

First we compute:

$$A^T A = \begin{bmatrix} 3 & 0 & 1 \\ -3 & 0 & 1 \end{bmatrix} \begin{bmatrix} 3 & -3 \\ 0 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 10 & -8 \\ -8 & 10 \end{bmatrix}$$

We then find the eigenvalues of this, so take:

$$A^T A - \lambda I = \begin{bmatrix} 10 - \lambda & -8 \\ -8 & 10 - \lambda \end{bmatrix}$$

So from this the characteristic equation of A becomes

$$(10-\lambda)^2 - 64 = 0$$

----> $$10 - \lambda = \pm 8$$

----> $$\lambda = 18, 2$$

So singular values of A are: $\sigma_1 = 3\sqrt{2}$ , $\sigma_2 = \sqrt{2}$

Now find associated eigenvectors for $A^TA$

$$A^TA - 18I = \begin{bmatrix} -8 & -8 \\ -8 & -8 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \text{ ----> eigenvector: } \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$A^TA - 2I = \begin{bmatrix} 8 & -8 \\ -8 & 8 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix} \text{ ----> eigenvector: } \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

So normalizing these, we take:

$$v_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ 1 \end{bmatrix} \quad , \quad v_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Those are what create the V matrix. So then for U,

$$u_1 = \frac{1}{\sigma_1} Av_1 = \frac{1}{3\sqrt{2}} \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} 3 & -3 \\ 0 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

---->

$$u_1 = \frac{1}{6} \begin{bmatrix} -6 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix}$$

And then:

$$u_2 = \frac{1}{\sigma_2} Av_2 = \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} 3 & -3 \\ 0 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

---->

$$u_2 = \frac{1}{2} \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

So we have

$$u_1 = \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix} \quad , \quad u_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Since $A$ is $3 \times 2$, $U$ must be $3 \times 3$, need to extend these to an orthonormal basis for $\mathbb{R}^3$

Can also note that these vectors are in $\mathbb{R}^3$ and the columns of $U$ need to be an orthonormal basis for the full space.

Fortunately, $u_1, u_2$ are essentially like standard basis elements so easy to find the third without really doing any computation, just take

$$u_3 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

So then we have U in full. Already found V, and $\Sigma$ is a diagonal matrix with singular values of A on its diagonal in decreasing order, yielding overall:

$$U = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}, \quad \Sigma = \begin{bmatrix} 3\sqrt{2} & 0 \\ 0 & \sqrt{2} \\ 0 & 0 \end{bmatrix}, \quad V = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix}$$

---

Might wonder if this representation is unique. Well $\Sigma$ uses singular values of A on its diagonal in decreasing order so no real variation in that. But U, what we picked for $u_3$ we easily could have chosen

$$u_3 = \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix}$$

and that would have worked just as well.

Somewhat surprising perhaps since making that change to $U$ doesn't alter $\Sigma$ or $V$, yet still works. But noting that that entry of $u_3$ only multiplies the last row of $\Sigma$ which is all 0s, it becomes clear why that's ok.

Also we took $v_1$ to correspond to $\begin{bmatrix} -1 \\ 1 \end{bmatrix}$ could quite naturally have done $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$ instead.

Unlike changing $u_3$, this change in $V$ will alter $U$, but still is a valid SVD on the whole.

So SVD breakdown not unique but not a ton of variation in it either.

This level of SVD problem I would consider reasonable for the exam. Ok such problems are never really short, but this is not excessively long either. A 3 x 2 or 2 x 3 matrix would be the most probable size for you to be given.

Does that mean a 2 x 3 would be roughly the same difficulty as 3 x 2? Well of course the specifics of the matrices matters, but you would surely guess so. That might not be as clear as you'd think though. In that circumstance $A^T A$ will be 3 x 3 not 2 x 2! And as you might recall doing diagonalization ideas (eigenvalues and eigenvectors) can be much harder in 3 x 3!

On other hand, in that case $V$ is 3 x 3 and $U$ is 2 x 2 so we'd be less likely to need to "add" more vectors to create the full $U$ since $V$ likely will provide enough.

In 3 x 2, $V$ can only provide two so you will definitely need at least one extra vector to create $U$.

In this problem we did we got kind of lucky with that as the two eigenvectors we automatically got for $U$ made it easy to fill out the remaining one. But that won't always be the case!

In short, while rough size certainly plays a role in how difficult the problem is likely to be, lots of small things and nuances beyond size which play a role as well.

2 x 2 and 3 x 3 also possible to be given for SVD but less likely since those can often use regular diagonalization which is easier to find and often more convenient of a form anyway.

But ok some square matrices not diagonalizable over $\mathbb{R}$ due to repeated or nonreal eigenvalue. For those SVD might be best option, especially if you don't wish to use imaginary numbers.

# Chapter B4

Now we are getting into differential equations (DEs) part of the class with occasional linear algebra ideas mixed in (hence why the crossover in this class despite the topics initially feeling totally disconnected).

Note that this is a totally different textbook than the linear algebra portion, your book is custom for Berkeley, puts them together. I denote the DEs chapters with the "B" in front to distinguish, they are not actually labelled as such.

Important to get back in mode of doing derivatives and integrals as we haven't really done those in this class prior. Ok, those ideas should have been done in Math 52 or the equivalent, but again might have been awhile, likely many folks could be a bit rusty at that kind of thing, probably many will need to review those a bit.

First section in this chapter is about application to force/mass. But we want to really focus on solving so skipping that and going into second section.

## B4.2: Second order linear differential equations

Say $y$ is a function of $t$, this is an equation of form

$$a(t)y'' + b(t)y' + c(t)y = f(t)$$

where $a(t), b(t), c(t), f(t)$ are any functions of $t$, just generally assume $a(t)$ is not identically 0 as otherwise this would really be first order. I.e. order refers to the highest derivative in the equation, so why the above is called second order.

Again these functions of $t$ can be any functions, exponential, trig, etc. So why is this called linear when those styles of functions are definitely not what we would categorize as linear usually?

Point is we can't have terms like $y^2$, $\sin y$, $e^y$
so it's linear in $y$ is the point really, not linear in $t$.

Remember we generally assume $a(t)$ not identically 0 to keep it second order. But how did we solve first order linear?

We used an *Integrating Factor.* That is, in

$$y' + P(t)y = Q(t)$$

multiply by

$$I(t) = e^{\int P(t)\,dt}$$

Good idea but very specific/delicate, how would you even think to do that in that circumstance?

Regardless of how, complicated enough that no realistic chance to expand that idea to second order linear. Just in general, second order is far harder to solve than

first order.

Mostly we can't think about solving second order linear in the general form. So much more difficult that we have to mostly try in a very restricted case, namely this style:

$$ay'' + by' + cy = f(t)$$

where $a, b, c$ are now constants, $a$ still nonzero.

This is a second order linear equation that has <u>constant coefficients.</u>

This is the type we will focus on solving as again nonconstant coefficients not really generally doable for us.

In calculus you did not do that either, other than when the coefficients were very simple nonconstants like say $t$ or $t^2$ style.

But even that we did by solving via series (likely the last thing you would have done in Math 52). And that's not necessarily super desirable, series is hard and the the answer will usually be a series itself, not a closed form so not very convenient to work with. We face the same challenges with that in this class.

But unlike in calculus we will deal with some nonconstant coefficient equations as some ideas we go over could make some very specific nonconstant coefficient equations accessible to us by ideas other than series.

---

Getting back to what we were doing, we wanted to solve:

$$ay'' + by' + cy = f(t)$$

But this is still a bit broad to start with as the form of $f(t)$ can greatly affect the calculation/answer. So make it simpler still: try to solve

$$ay'' + by' + cy = 0$$

That is, assume $f(t) = 0$. This is called <u>homogeneous</u>, and with the $f(t)$ still there, and that not being identically zero, it's called <u>nonhomogeneous.</u>

Is this similar in dyanmics to the matrix equations $Ax = 0$ vs. $Ax = b$ we studied early in course? Well uses the same terminology so we suspect it likely is!

How did we solve matrix equation $Ax = b$? Well one way was to solve $Ax = 0$, find its general solution, then find one particular solution to $Ax = b$. We can do same here as it turns out, but the derivatives involved certainly change the dynamics.

We'll see that play out later, but we first need to solve the homogeneous equation

$$ay'' + by' + cy = 0$$

How? Well when solving the similar first order counterpart $y' = ky$, exponential growth/decay, the general solution was $y = Ce^{kt}$

So turns out that is logical thing to try here as well.

Also logical since all these terms same style, seems to make sense as a way to satisfy this, moreso than say $y = \sin t$ since that doesn't stay completely consistent in how derivatives look. So try

$$y = Ce^{rt} \quad \text{----}> \quad y' = Cre^{rt} \quad \text{----}> \quad y'' = Cr^2e^{rt}$$

Plugging these into equation and simplifying gives

$$e^{rt}[ar^2 + br + c] = 0$$
$$ar^2 + br + c = 0$$

This is called the <u>characteristic equation</u> (another crossover term with linear algebra!)

Like any quadratic equation, this has two complex solutions.

The different possibilities (which affect how this works out) are:

(1) Two distinct real solutions
(2) One repeated real solution
(3) Two nonreal solutions (which are complex conjugates).

Each of these is handled a bit differently. Again, not dissimilar to when we were try to diagonalize 2 x 2 real matrices, each of the above natures for the roots of the characteristic polynomial had to be handled a bit differently.

This section focuses on (1) + (2).

---

We generally had two style of problems in this when we were going to decode full solution, i.e. solve for the unknown constants that arise from the integration.

Those were:

(1) Initial value problem: $y(t_0) = y_0$ , $y'(t_0) = y_1$

(2) Boundary value problem: $y(t_0) = y_0$ , $y'(t_1) = y_1$

First gives function and derivative value at same point. Latter gives function at two distinct points.

Interesting fact is that, regardless of values, initial value problem always has unique solution while boundary value problem can have no solution, unique solution, or infinite solutions (just like linear system - hence you see more similarities!).

Why does initial value problem always have a unique solution? Might seem surprising that that's always the case no matter the point or values chosen.

Say we have the case where characteristic equation has two distinct real solutions, say $r_1, r_2$. Then we have general solution:

$$y(t) = c_1 e^{r_1 t} + c_2 e^{r_2 t}$$

so that

$$y'(t) = c_1 r_1 e^{r_1 t} + c_2 r_2 e^{r_2 t}$$

Plugging in the initial value problem conditions and thinking of it as a linear system, after simplification we would get a coefficient matrix of

$$\begin{bmatrix} 1 & 1 \\ r_1 & r_2 \end{bmatrix}$$

And this is invertible since $r_1 \neq r_2$ . That's key. Remember in linear system with square coefficient matrix, if that matrix is invertible, there is always a unique solution, regardless of what you are solving equal to.

Hence again you see linear algebra ideas play into this.

Next time will clarify these issues as well as talk more about these types of equations in the different situations based on the style of roots we wind up with from the characteristic equation.